



Semester One Examination, 2024

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section One: Calculator-free

If required by your examination administrator, please place your student identification label in this box

WA student number: In figures

--	--	--	--	--	--	--	--

In words

--	--	--	--	--	--	--	--

Circle your Teacher's Name: Mr Bartoshefski Mrs Fraser-Jones Ms Horley Dr Kang
Ms Lastari Ms Nathan Mr Singh

Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Number of additional
answer booklets used
(if applicable):

--

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	10	10	100	98	65
Total					100

Instructions to candidates

- The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
- You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
- Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
- It is recommended that you do not use pencil, except in diagrams.
- Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
- The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
1	4	
2	8	
3	10	
4	8	
5	7	
6	8	
7	7	
Section 1 Total	52	
Weighted ($\times 0.6731$)	35%	
Section 2 Weighted	65%	
Total	100%	

Section One: Calculator-free**35% (52 Marks)**

This section has **seven** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Question 1

(4 marks)

- (a) Determine the value of the expression $\frac{15}{x^2 - \sqrt{y}}$ when $x = 4$ and $y = 36$. (2 marks)

- (b) Use the formula $A = \frac{h}{2}(a + b)$ to determine the value of A when $a = 15$, $b = 27$ and $h = 10$. (2 marks)

Question 2**(8 marks)**

- (a) An item is priced at \$135 excluding GST. Determine the price of the item once 10% GST has been included. (2 marks)
- (b) A homeowner is offered a 2% discount for early payment of their \$1700 annual council rates bill. How much would they pay if they took up the offer? (2 marks)
- (c) A watch enthusiast paid \$500 for a timepiece but later sold it for just \$440. Determine the percentage loss made by the enthusiast. (2 marks)
- (d) A gallery charges a 35% commission on the price of all artworks that it sells. If it sells a small sculpture for \$560 what commission will it charge? (2 marks)

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Question 3

(10 marks)

Use these four matrices to answer the following questions.

$$\mathbf{A} = \begin{bmatrix} 5 & -2 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 2 & 6 \\ 5 & -4 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 4 & 6 \\ 5 & -2 \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$$

- (a) Two of the following matrix calculations are **not** possible.

$$\mathbf{AB} \quad \mathbf{BC} \quad \mathbf{BA} \quad \mathbf{B} + \mathbf{C} \quad \mathbf{C} + \mathbf{D}$$

Circle the two that are not possible and explain why.

(4 marks)

- (b) Determine $\mathbf{C} - 2\mathbf{B}$.

(2 marks)

- (c) Determine $\mathbf{CB}$.

(2 marks)

- (d) Determine a matrix $\mathbf{P}$ which, when added to $\mathbf{B}$, results in a zero matrix.

(2 marks)

Question 4

(8 marks)

- (a) A free veterinary clinic at a show treated 11 skin, 8 stomach and 5 other ailments on Saturday and the following day treated 7 skin, 6 stomach and 9 other ailments. Express the numbers for Saturday in row matrix A , do the same for the Sunday numbers in row matrix B , and then determine matrix $A + B$ that represents the numbers of each type of ailment treated over the weekend. (2 marks)

- (b) A stall at the same event on Saturday sold small show bags for \$15 each and large show bags for \$20 each. The sales for this stall are shown in the table below.

	Small	Large
First hour	6	11
Second Hour	10	16

- (i) Represent the number of bags sold in the 2×2 matrix N . (1 mark)

- (ii) Represent the costs of a small and a large show bag in matrix C so that the matrix product NC will produce a meaningful result. (1 mark)

- (iii) Determine the matrix $X = NC$, and explain what the entry $x_{1,1}$ represents. (3 marks)

- (iv) Determine the stall's total income from the sale of show bags during the first two hours of the event. (1 mark)

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Question 5**(7 marks)**

Information about two companies listed on a stock exchange is shown in the table below.

Company Code	Share price (\$)	Annual earnings per share (cents)	Last dividend paid per share (cents)
CBQ	1.44	3	5
QLM	2.00	5	7

- (a) Determine the last dividend payment to an investor who owned 4200 shares in CBQ. (2 marks)
- (b) The last dividend paid per share by QLM was 7 cents when its share price was \$2.00. Determine the equivalent percentage dividend. (2 marks)
- (c) Calculate the price-to-earnings ratio for each company. (2 marks)
- (d) State, with justification, in which of the two companies an investor would pay the least for each dollar of the company's earnings. (1 mark)

Question 6

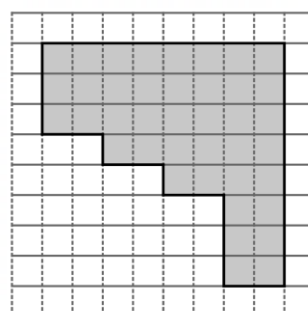
(8 marks)

- (a) A floor plan of a home is drawn where 4 cm on the plan represents 2.4 m in the actual home. The scale factor for this plan is $1:n$, where n is a whole number. Determine the value of n . (2 marks)

- (b) The side lengths of triangle A are 5 cm, 7 cm and 8 cm. The side lengths of triangle B are 20 cm, 28 cm and 36 cm. Are these triangles similar figures? Justify your answer. (3 marks)

- (c) The shaded region on the grid to the right is a scale drawing of a concrete slab, where each small square has a side length of 4 mm.

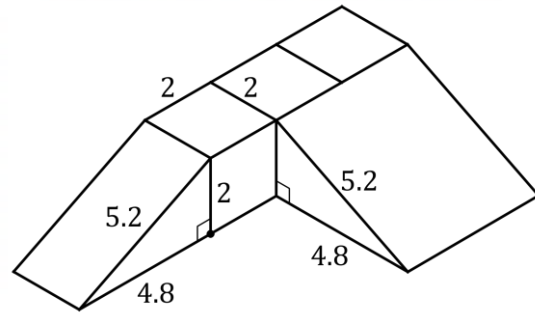
If the area of the actual concrete slab is 360 square metres, determine the scale factor of the drawing.



(3 marks)

Question 7**(7 marks)**

A large three-dimensional sculpture is to be suspended outside a sports arena to advertise a skateboarding competition. The sculpture comprises three two-metre cubes and two triangular prisms that will be glued together to form the composite figure as shown in the diagram (not to scale).



All dimensions are in metres and the triangular cross-section of each prism has sides of 5.2 m, 4.8 m and 2 m.

- (a) Show that the volume of the smaller triangular prism is 9.6 cubic metres. (1 mark)
- (b) Determine the volume of the composite figure. (2 marks)
- (c) The sculpture needs to be coated with a waterproof sealant spray. Determine the total surface area of the composite figure that needs to be covered. Show clear evidence of the method you use. (4 marks)

Supplementary page

Question number: _____

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

